Multihop Routing for IoT-Based Digital Twin: Novel Metaheuristic Approaches

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***Abstract*—This article addresses the challenge of optimizing multihop routing in Internet of Things (IoT)-based digital twin systems, referred to as the MOUNTAIN problem. Multihop routing is inherently complex due to the need to balance energy consumption and communication reliability across multiple nodes, especially in dynamic and large-scale IoT networks. In MOUNTAIN, multiple IoT devices in the physical network (PN) frequently transmit data to the digital network twin (DNT), managed by a central server. Given the limited energy resources of IoT devices, our approach considers both energy efficiency and communication reliability. We formulate the MOUNTAIN problem as an optimization task aimed at reducing overall energy consumption while maintaining robust data transmission. Moreover, we address the problem with both single-task optimization and multitask optimization and propose two corresponding evolution-based metaheuristics that utilize well-designed solution representations and genetic operators to obtain near-optimal solutions to the problem. Among them, the proposed single-task evolutionary algorithm (STEA) solves each problem instance independently, while the proposed multitask evolutionary algorithm (MTEA) solves multiple instances at the same time to take advantage of exchanging useful solution information during parallel solution searches. Extensive experiments on synthetic datasets demonstrate that our proposed algorithms significantly outperform existing methods, reducing energy consumption and improving network stability. This research contributes to the development of sustainable and efficient IoT infrastructures, which are essential for the operational demands of digital twin applications.**

***Index Terms*—Digital twin, energy minimization, evolutionary algorithm, Internet of Things (IoT), multihop routing.**

I. Introduction

THE METAVERSE represents a transformative convergence of physical and digital realms through immersive technologies, with digital twins—virtual replicas of physical entities—serving as a critical foundation. These digital twins rely on data collected from Internet of Things (IoT) devices to maintain accurate real-time representations [1], [2]. As the Metaverse evolves, the effectiveness of digital twins increasingly depends on the reliability and efficiency of the underlying IoT infrastructure that collects and transmits critical data.

A significant challenge in IoT-based digital twin systems is establishing robust communication pathways between distributed IoT devices and central data processing points. Many IoT devices operate in environments where direct communication with base stations (BSs) is unfeasible due to distance limitations or power constraints [3]. In such scenarios, multihop communication becomes essential, where data is relayed through intermediate devices before reaching its destination [2], [4]. However, this approach introduces complex optimization challenges related to energy consumption, bandwidth utilization, and system resilience that directly impact digital twin performance.

Despite extensive research on IoT routing protocols, existing approaches have notable limitations when applied to digital twin contexts. Current algorithms primarily focus on conventional metrics, such as latency or throughput without considering the unique requirements of digital twin synchronization [5], [6]. While recent studies like Yang et al. [7] have explored communication optimization for digital twins, they have overlooked the critical aspects of multihop routing and energy consumption minimization. Meta-heuristic optimization approaches offer promising solutions to these challenges due to their ability to efficiently search large solution spaces and handle complex constraints [8], yet their application to IoT-based digital twin systems remains largely unexplored.

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This article addresses the identified research gap by investigating multihop routing optimization in IoT-based digital twin systems and proposing novel meta-heuristic algorithms specifically designed to minimize energy consumption while maintaining reliable data transmission. Our approach not only advances the theoretical understanding of routing optimization in digital twin contexts but also provides practical solutions that can enhance the performance and sustainability of real-world IoT deployments supporting digital twin applications.

In summary, our main contributions are as follows.

1) We are the first to investigate the multihop routing in IoT-based digital twin systems, named *MOUNTAIN*, aimed at optimizing energy consumption in digital twin system, where physical devices can transmit data directly to one another and to a server in physical IoT network, enhancing network communication capabilities and stability of IoT-based digital twin systems.
2) We propose two novel meta-heuristic algorithms, specifically a single-task evolutionary algorithm (STEA) and a multitask evolutionary algorithm (MTEA) to efficiently solve the MOUNTAIN problem. Our approach features well-designed solution representations for MOUNTAIN, supported by theoretical analysis. Genetic operations, such as crossover, mutation, selection, and fitness evaluation, are designed to align with the efficient solution encoding and effectively search for good solutions to MOUNTAIN.
3) We conducted extensive experiments on various synthetic datasets to evaluate the effectiveness of our proposed algorithm against robust baseline methods, including the African vultures optimization algorithm (AVOA), as well as greedy and random search strategies. The results demonstrate that our algorithm consistently outperforms these baselines across all datasets and experimental scenarios, achieving superior performance within acceptable time limits. This underscores its effectiveness and suitability for addressing the MOUNTAIN problem.

The remainder of this article is organized as follows. Section II provides a comprehensive review of related work. Section III introduces the system model and problem formulation for the IoT-based digital twin system. Section IV describes the proposed evolutionary algorithms, while Section V presents the performance evaluation and simulation results. Finally, Section VI concludes this article. Frequently used notations are listed in Table I.

II. Related Work

This section presents a comprehensive review of relevant literature in the areas of digital twin technologies, IoT networks, multihop routing strategies, energy optimization, and meta-heuristic approaches for network optimization problems.

A. Digital Twin Technologies and IoT Integration

Digital twins have emerged as a transformative technology that creates virtual replicas of physical systems, enabling real-time monitoring, analysis, and optimization [9], [10]. Recent research has highlighted the critical role of IoT devices in enhancing the functionality and synchronization of digital twins within the Metaverse [2], [11]. Han et al. [2] proposed a dynamic hierarchical framework that facilitates IoT-assisted synchronization of digital twins, emphasizing the integration of real-time data from numerous IoT sensors. This framework addresses challenges related to data consistency and real-time interactions, enabling more seamless user experiences in virtual environments by ensuring that physical-world data is accurately reflected in digital twins.

Luong et al. [12] explored energy optimization in UAV-assisted vehicular Metaverse systems, where UAVs, acting as IoT devices, transmit data to a BS that supports a digital twin system. This setup enhances resource allocation and synchronization between UAVs and the digital environment, highlighting how strategic deployment of IoT devices, including UAVs, can improve system performance and ensure efficient communication between physical entities and their digital counterparts. The integration of edge computing, as highlighted by Yu et al. [13], imposes the need for a decentralized approach to data processing, enabling efficient data collection and analysis in real time, thereby bridging the gap between the physical and digital realms. These studies collectively emphasize the significance of a robust IoT infrastructure that can reliably transmit data to BSs, supporting the operational demands of digital twins and advancing the overall functionality of Metaverse applications.

The evolution of digital twin technologies has transitioned from simple virtual representations to complex, data-driven models that require continuous updates from distributed IoT sensors. This evolution has created a critical need for efficient data collection and transmission mechanisms that current digital twin frameworks did not adequately address. Despite the significant advancements in digital twin modeling and synchronization techniques, there remains a substantial gap in research concerning the optimization of the underlying communication infrastructure, particularly in scenarios requiring multihop data transmission across resource-constrained IoT networks.

B. Multihop Routing in IoT Networks

Various studies have explored the deployment strategies and challenges of IoT devices within these contexts, emphasizing their role in enhancing data fidelity for the Metaverse and digital twins [2], [9], [12], [14]. However, several IoT devices operate in remote or hard-to-reach areas, posing connectivity and resource challenges. These devices often cannot connect directly to central data fusion points, such as BSs, due to their location or limited access to bandwidth and energy resources. As a result, they rely on nearby IoT devices with greater resources to forward data efficiently.

To address this, remote IoT devices employ multihop communication, utilizing nearby devices with sufficient resources, such as bandwidth and energy, to forward data to the BS. Multihop communication is well established in wireless sensor networks (WSNs) as a strategy for expanding network coverage and efficiently managing limited resources [2], [3], [4].

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TABLE I
LIST OF NOTATIONS FREQUENTLY USED IN THIS ARTICLE

Notation	Description
a_n	Accuracy of the system at each time slot n
A_n	Accuracy requirement at time slot n
B	Bandwidth of each resource block
C_{nk}	Number of devices transmitting data to device k at time slot n
d_{nk}	Transmitted data of device k at time slot n
D_{nk}	Generated data of device k at time slot n
eda_{nk}	Energy consumption of device k for data aggregation
e_{nk}	Total energy consumption of device k at time slot n
er_{nk}	Energy consumption of device k for receiving data
et_{nk}	Energy consumption of device k for transmitting data
m	Waterfall threshold
M	Number devices equipped with relay capabilities
$\mathcal{M} = \{1, \cdots, M\}$	Set of devices with relay capabilities
N	Number of time slots in a long period T
$\mathcal{N} = \{1, \cdots, N\}$	Set of time slots
N_0	Noise power spectral density
K	Number of IoT devices
$\mathcal{K} = \{1, \cdots, K\}$	Set of devices
$k \in \mathcal{K}, n \in \mathcal{N}$	Indexes of physical device and time slot, $\lvert\mathcal{K}\rvert = K, \lvert\mathcal{N}\rvert = N$
K_0	Maximum number of resource blocks
h_{nk}	Channel gain between physical device k and the server at time slot n
p_{nk}	Transmit power of device k at time slot n
P_n	Maximum transmit power at time slot n
r_c	Communication range
r_{nk}	Transmission rate between device k and the server at time slot n
t_n	Transmission delay of physical network at time slot n
t_{nk}	Wireless transmission delay of device k at time slot n
T_0	Duration of each time slot calculated by $\frac{T}{N}$
u_{kq}	Euclidean distance between two devices k and q
$x_{nk} \in \{0, 1\}$	Decision variable indicates whether device k transmits data to BS
$\mathcal{V}$	Set of physical devices, including BS and K IoT devices
$\mathcal{E}$	Set of connections (links) between devices

This approach enables robust data transmission pathways where direct access is unfeasible, promoting energy conservation and balancing loads across the network [1].

The integration of digital twin technology with IoT systems presents an unprecedented opportunity to enhance real-time monitoring and control capabilities across various applications. However, to fully leverage the potential of digital twins, it is essential to address the challenges associated with multihop routing in IoT networks. Multihop routing enables efficient communication among distributed sensor nodes, which is crucial for maintaining the accuracy and reliability of digital twin systems. Recent studies, such as those by Fan and Xin [5], have explored clustering methods in this context. However, it remains unclear whether these approaches adequately incorporate digital twin paradigms, necessitating a thorough investigation of existing multihop routing literature.

A review of recent works in multihop routing reveals significant shortcomings, particularly in the context of modern IoT systems that integrate digital twin frameworks. Many of these studies fail to address the unique requirements of such systems, limiting their applicability and effectiveness in real-world applications. Notable examples include Fan and Xin [5], who proposed a clustering and routing algorithm for large-scale WSNs without considering energy efficiency in the context of digital twin integration, and Kaur et al. [6], who introduced an intelligent routing scheme but do not address the challenges of energy consumption in IoT-enabled systems.

Furthermore, Chen et al. [15], Shen et al. [16], and Nguyen et al. [17] employed traditional routing protocols that neglect energy efficiency and the integration of digital twins, while Chithaluru et al. [18], Ojha and Chanak [19], and Prajapat et al. [20] explored energy-efficient routing, but failed to account for the computational requirements and dynamic synchronization needed in IoT networks with digital twin frameworks. These limitations underscore the need for novel routing algorithms that not only minimize energy consumption but also integrate digital twin models for more accurate, real-time data processing and decision-making in IoT systems.

C. Energy Optimization in IoT Systems

Energy efficiency is paramount in IoT networks, as most devices operate on limited power sources, such as batteries or energy harvesting systems. The longevity and reliability of IoT-based digital twin systems fundamentally depend on effective energy management strategies that extend network lifetime while maintaining adequate data collection capabilities [21], [22], [23].

Current approaches to energy optimization in IoT networks can be classified into three main categories: 1) hardware-level optimizations; 2) MAC-layer strategies; and 3) network-layer techniques [24], [25]. While hardware and MAC-layer optimizations provide important foundational improvements, network-layer techniques—particularly routing optimizations—offer the greatest potential for system-wide energy efficiency enhancements.

Rosabal et al. [26] investigated worst case energy consumption in UAV-assisted IoT networks, developing models to predict and minimize energy usage. However, their approach focuses primarily on UAV energy optimization rather than addressing the comprehensive energy management required across heterogeneous IoT networks supporting digital twins. Similarly, Ghorbel et al. [4] optimized energy efficiency in wireless data gathering using UAVs but did not consider the continuous operation requirements of digital twin systems.

Recent research by Yu et al. [13] highlighted the importance of resource management techniques for reliable communication in resource-constrained environments. While these methods address connectivity challenges, they often fail to consider the intricate relationship between routing decisions and energy consumption patterns specific to digital twin data flows.

Yang et al. [7] presented a framework for joint communication and computation optimization for digital twins, but notably overlooked energy consumption considerations. This oversight is significant because energy constraints frequently serve as the limiting factor in IoT network performance and longevity. Without accounting for energy efficiency, even the most sophisticated communication and computation optimization approaches will ultimately fail in real-world IoT deployments supporting digital twins.

D. Meta-Heuristic Approaches for Network Optimization

Network optimization problems, particularly those involving routing in multihop wireless networks, are typically NP-hard and characterized by large solution spaces with complex constraints. Metaheuristic algorithms, particularly those inspired by natural and evolutionary processes [8], [27], [28], [29], [30], [31], have been effectively applied to these complex problems that traditional methods struggle to solve. These

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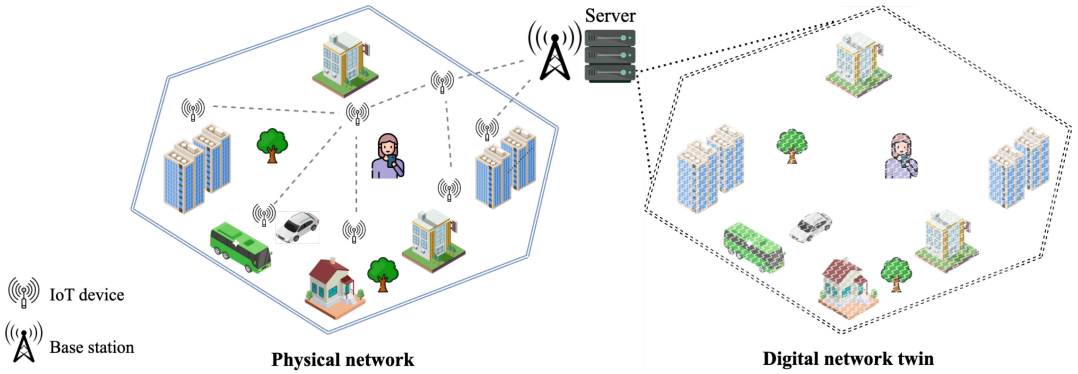


Fig. 1. Considered digital twin system model based on IoT network.

algorithms operate on a population of solutions, utilizing various operators to iteratively search for optimal outcomes. The design of appropriate encoding schemes and specialized operators is crucial, as they enhance the algorithm's ability to explore and optimize the search space efficiently.

In recent years, multitasking evolutionary algorithms have incorporated knowledge transfer mechanisms to optimize multiple tasks simultaneously [32], [33], [34], [35]. This approach involves sharing information between tasks to accelerate convergence across all optimization objectives. Notably, multifactorial evolutionary algorithms (MFEAs) have been employed in network design problems [36], [37], [38], [39], [40], focusing on network lifetime optimization in WSNs.

These metaheuristic approaches have demonstrated promising results in addressing complex optimization problems in wireless networks. However, their application to the specific challenges of IoT-based digital twin systems, particularly with respect to multihop routing and energy optimization, remains an area with significant potential for exploration and innovation. The integration of digital twin technologies to enhance energy efficiency and overall performance in IoT networks presents an unexplored yet promising research direction that our work aims to address.

III. System Model and Problem Formulation

We define the MOUNTAIN in the following sections.

A. System Model and Assumptions

We consider the digital network twins (DNTs) that include an IoT network and a mapping DNT that is generated and controlled by a central server [7] as illustrated in Fig. 1. Whereas the physical network (PN) (IoT network) includes K IoT devices as physical devices and one BS (server). These physical devices must transmit status data to the central server, which uses the data to generate DNT. In this article, it is important to highlight that devices are capable of both connecting to and transmitting data directly to one another. We consider an IoT network consisting of M ($M < K$) devices equipped with relay capabilities. These devices not only receive data from other devices within the network but also have the ability to forward this data to the central server. This relay functionality enhances the efficiency and robustness of the data transmission process within the network. We consider a time period denoted as T, which is divided into N time slots. The duration of each time slot is calculated as $T_0 = (T/N)$.

At time slot n, the device k in the PN generates data D_{nk}. The generated data can be transmitted directly to the server or routed through another relay device. To ensure synchronization between the PN and the DNT, the physical device k can select to transmit data d_{nk} (in bits) to the server. The transmission rate from the device k to the server is defined as follows:

$$r_{nk} = B \log_2\left(1 + \frac{p_{nk} h_{nk}}{N_0 B}\right) \tag{1}$$

where B represents the bandwidth of each resource block assigned to device k, p_{nk} denotes the transmit power of device k at time slot n, h_{nk} is the channel gain from physical device k to the server, and N_0 indicates the noise power spectral density. Particularly, the channel gain h_{nk} consists of Rayleigh fading channel gain and long-term path loss with shadowing. Due to the randomness of the channel gain, the data received at the server may contain errors. The transmission error rate at time slot n can be expressed as follows [15]:

$$te_{nk}(p_{nk}) = 1 - e^{-\frac{mN_0 B}{p_{nk} h_{nk}}} \tag{2}$$

where m is a waterfall threshold.

As the transmitted data cannot exceed the generated data, we have

$$\sum_{i=1}^{n} d_{ik} c_{ik}(p_{ik}) \le \sum_{i=1}^{n} D_{ik} \quad \forall n \in \mathcal{N}, k \in \mathcal{K} \tag{3}$$

where

$$c_{ik}(p_{ik}) = \begin{cases} 1, & \text{with probability } e^{-\frac{mN_0 B}{p_{ik} h_{ik}}} \\ 0, & \text{with probability } 1 - e^{-\frac{mN_0 B}{p_{ik} h_{ik}}} \end{cases}. \tag{4}$$

In each time slot, let $x_{nk} \in \{0, 1\}$ represent whether the device k in the PN is transmitting data to the server. The notation $x_{nk} = 1$ indicates that physical device k is transmitting data to the BS; otherwise, $x_{nk} = 0$.

This article considers frequency-division multiple access (FDMA) for uplink transmission. Due to the limited number of resource blocks in the communication system, the maximum

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number of devices that can be associated in each time slot is restricted, i.e.,

$$\sum_{k=1}^{M} x_{nk} \leq K_0 \quad \forall n \in \mathcal{N} \tag{5}$$

with K_0 being the maximum number of resource blocks that can be utilized in the communication system.

The energy consumption of a device k when it receives d_{np} bits of data from device p can be calculated as [40]

$$er_{nk} = d_{np} \times \epsilon_{elec} \tag{6}$$

where ϵ_{elec} is the energy consumed by the transceiver circuit or receive circuit and is commonly set to 50 nJ/bit.

The energy consumption of a device k to aggregate d_{np} bits of data from device p is calculated as [40]

$$eda_{nk} = d_{np} \times \epsilon_{DA} \tag{7}$$

where ϵ_{DA} is the energy consumed for aggregating data and is set to 5 pJ/bit.

The energy consumption for each device k to transmit d_{nk} bits data to device q at each time slot n can be calculated as [40]

$$et_{nk} = d_{nk} \times \left(\epsilon_{elec} + \epsilon_{fs} \times u_{kq}^2\right) \tag{8}$$

where ϵ_{fs} is the energy expenditure for transmitting d_{nk} bits of data to achieve an acceptable bit error rate for the free space model. The value of this parameter is 10 pJ/bit/m^2 [40]. u_{kq} is the Euclidean distance between devices k and q.

Therefore, the energy consumption for each device k to receive, gather, and transmit data from C_{nk} devices to the device q (can be a BS) at each time slot n is given in

$$\begin{aligned} e_{nk} &= er_{nk} + eda_{nk} + et_{nk} \\ &= \sum_{p=1}^{C_{nk}} d_{np} \times \epsilon_{elec} + \left(d_{nk} + \sum_{p=1}^{C_{nk}} d_{np}\right) \times \epsilon_{DA} \\ &\quad + \left(d_{nk} + \sum_{p=1}^{C_{nk}} d_{np}\right) \times \left(\epsilon_{elec} + \epsilon_{fs} \times u_{kq}^2\right) \\ &= \sum_{p=1}^{C_{nk}} d_{np} \times \left(2\epsilon_{elec} + \epsilon_{DA} + \epsilon_{fs} \times u_{kq}^2\right) \\ &\quad + d_{nk} \times \left(\epsilon_{DA} + \epsilon_{elec} + \epsilon_{fs} \times u_{kq}^2\right). \end{aligned} \tag{9}$$

At each time slot n, the wireless transmission delay of a device k can be calculated as follows [7]:

$$t_{nk} = \frac{e_{nk}}{p_{nk}}. \tag{10}$$

As multiple devices can communicate with the server simultaneously, the transmission delay of the PN during time slot n is determined by the maximum transmission delay among all devices

$$t_n = \max_{k \in \mathcal{K}} x_{nk} t_{nk}. \tag{11}$$

The accuracy of the DNT relies on obtaining more data from the PN. However, accumulating large amounts of data can result in increased transmission delays and energy consumption. Therefore, it is crucial to explore the tradeoff between accuracy and the wireless costs associated with the PN, which include both delay and energy consumption. In our system, the cumulative transmitted data is defined as

$$P = \sum_{i=1}^{n} \sum_{k=1}^{K} d_{ik}\, c_{ik}(p_{ik}) \tag{12}$$

and the cumulative generated data as

$$Q = \sum_{i=1}^{n} \sum_{k=1}^{K} D_{ik}. \tag{13}$$

The ratio $R = (P/Q)$, $(0 \leq R \leq 1)$ represents the fraction of generated data that is successfully transmitted. To account for nonlinear performance effects, we introduce a sensitivity parameter $\alpha > 0$ and define the accuracy metric following [7] as:

$$a_n = f(P, Q) = \left(\frac{P}{Q}\right)^{\alpha}. \tag{14}$$

When $alpha = 1$, the metric is linear; if $\alpha > 1$, small improvements in R at low values result in a larger increase in a_n, whereas $0 < \alpha < 1$ yields a more gradual improvement.

In real-world scenarios, IoT devices continuously generate data and transmit it to a central server, with successful transmission verified by error-detection methods (e.g., checksums or acknowledgments) [41] that determine $c_{ik}(p_{ik})$. The server aggregates P and Q, computes R, and then derives a_n. The parameter α is calibrated using empirical or simulation data to reflect the system's performance sensitivity. Moreover, smoothing techniques (such as moving averages) are applied to mitigate transient fluctuations, providing a stable accuracy measure that informs adaptive routing and power allocation strategies.

B. Problem Formulation

We consider an IoT network represented as a connected undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{s_0, s_1, \ldots, s_K\}$ represents the set of physical devices and $\mathcal{E}$ represents the set of links among the devices. The BS is denoted as s_0, and the kth physical device is denoted as s_k $(k = 1, \ldots, K)$. In the IoT network, two devices have connectivity if the distance between them is within the communication range r_c. We then formalize the MOUNTAIN problem as follows.

Input:

1) An IoT network as in the system model represented by a connected and undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. $\mathcal{V}$ is a set of physical devices, and $\mathcal{E}$ is a set of links between the devices.
2) The location of physical devices (2-D coordinates), including the BS s_0, and physical device $s_1, \ldots, s_K$.
3) The generated data of each physical device, denoted as d_{nk} (in bits), can be transmitted to the BS. The initial

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transmit power of the kth device at time slot n is denoted as p_{nk}.

Output:

1) A routing of physical devices in each time slot is represented by a spanning tree of the input graph G, i.e., $\mathcal{T} = (\mathcal{V}, \mathcal{E}_{\mathcal{T}})$ rooted at s_0, where $\mathcal{E}_{\mathcal{T}} \subset \mathcal{E}$ and $|\mathcal{E}_{\mathcal{T}}| = K + 1$.
2) A set of decision variables $X = (x_{nk})_{N\times K}$, where $x_{nk} = 1$ if s_k is directly connected to the server s_0 in the time slot n.

Constraints:

1) The accuracy of each time slot must be larger than or equal to the accuracy lower bound requirement A_n

$$f\left(\sum_{i=1}^{n}\sum_{k=1}^{K} x_{ik}d_{ik}c_{ik}(p_{ik}), \sum_{i=1}^{n}\sum_{k=1}^{K} D_{ik}\right) \geq A_n \ \forall n \in \mathcal{N}. \tag{15}$$

2) The latency (equivalent to delay) of each time slot must not exceed the maximum latency permission L_n

$$\max_{k\in\mathcal{K}} x_{nk}t_{nk} \leq L_n \quad \forall n \in \mathcal{N}. \tag{16}$$

3) Each physical device should have regular communication with the server at least $\beta_k\tau$ times within any time window of length τ

$$\sum_{i=n}^{n+\tau} x_{ik} \geq \beta_k\tau \quad \forall n \in \mathcal{N}, k \in \mathcal{K}. \tag{17}$$

Here, $\beta_k \in (0, 1]$ is a parameter specific to device k, indicating the proportion of time slots in which communication must occur. τ is a time window defining the number of slots to evaluate.

4) The number of devices directly connected to the server cannot exceed the maximum number of resource blocks K_0

$$\sum_{k=1}^{M} x_{nk} \leq K_0 \quad \forall n \in \mathcal{N}. \tag{18}$$

5) The wireless transmission delay of each device cannot exceed the time frame of each slot T_0

$$0 \leq t_{nk} \leq T_0 \quad \forall n \in \mathcal{N}, k \in \mathcal{K}. \tag{19}$$

Objective:

1) Due to the energy constraints of the IoT devices, the objective is to minimize the total energy consumption of the system

$$\min \sum_{n=1}^{N}\sum_{k=1}^{K} e_{nk}. \tag{20}$$

IV. Methodology

The MOUNTAIN above is a problem with a complex solution search space due to searching for routing trees on a large input graph. Furthermore, this problem has many complex constraints, e.g., nonconvex, and representing the solution to satisfy these constraints requires a meticulous and appropriate design. Therefore, we propose two metaheuristics based on evolutionary algorithms using efficient solution encoding and search operators to solve this MOUNTAIN problem effectively. Specifically, we first introduce an STEA, using node-depth encoding to represent routing trees to help minimize the solution search space. We second present an MTEA, a recent advanced state-of-the-art evolutionary algorithm to solve MOUNTAIN by using a multitask optimization approach. In addition, effective genetic operators, including crossover and mutation, help create good solutions that are always valid and satisfy the constraints of the problem without the extra effort of modifying the solution.

Algorithm 1: STEA for MOUNTAIN

```
1  begin
2      t ← 0;
3      /* initialization stage                                   */
4      P(0) ← Initialize L valid individuals randomly;
5                  ▷Refer to subsection IV-B ;
6      /* improvement stage                                      */
7      while terminating condition is not met do
8          Offspring population P_c(t) ← ∅;
9          while |P_c(t)| < L do
10             p_a, p_b ← Randomize two parents from P(t);
11             /* p_c, p_m is the probability of crossover and
                  mutation, respectively */
12             if (rand(0, 1) < p_c) then
13                 o ← Perform crossover ECO(p_a, p_b); ▷See
                     IV-C ;
14             if (rand(0, 1) < p_M) then
15                 o ← Perform mutation EPO(o); ▷See IV-C ;
16             P_c(t) ← P_c(t) ∪ o;
17         end
18         Construct a solution and evaluateÂăit for each child in
             P_c(t);
19         P(t) ← P(t)∪P_c(t);
20         /*Survival selection*/ P(t + 1) ← Select L best
             individuals in P(t);
21         t ← t + 1;
22     end
23 end
```

The overall framework of our STEA and MTEA is described in Sections IV-A and IV-E. The general solution representation for MOUNTAIN is presented in Section IV-B. We develop genetic operators, i.e., recombination and a mutation operator, suitable for MOUNTAIN shown in Section IV-B. The fitness evaluation for the solution of MOUNTAIN is described in Section IV-D. Last but not least, the time complexity of the proposed algorithms is analyzed in Section IV-F.

A. STEA: Proposed Single-Task Evolutionary Algorithm

In the STEA, we tackle each MOUNTAIN instance independently. The overall structure of STEA is outlined in Algorithm 1. Our algorithm consists of two main stages: 1) initialization (lines 2–4) and 2) improvement (lines 5–20). In the initialization stage, we generate the first population of L individuals by constructing L random spanning trees from the input graph. In the improvement stage, we produce the offspring population using two genetic operators: 1) recombination and 2) mutation (lines 6–15). The offspring

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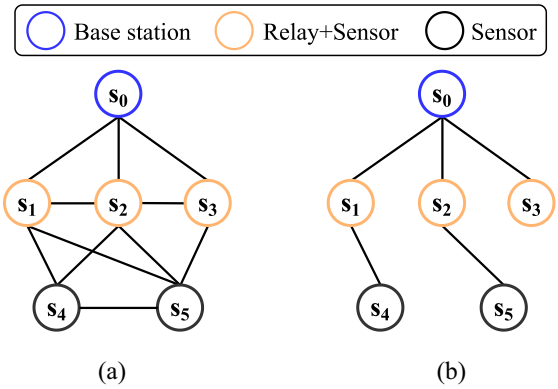


Fig. 2. Example of an IoT network represented by a graph and a possible routing tree solution for the MOUNTAIN problem. (a) Input graph. (b) Routing solution.

Node	s_0	s_1	s_4	s_2	s_5	s_3
Depth	0	1	2	1	2	1

Encoding (genotype)

Fig. 3. Example of solution encoding for our method to solve the MOUNTAIN problem.

individuals are then decoded into solutions, and their fitness is evaluated (line 16). After evaluating fitness, the offspring population is merged with the current population. We apply Elitism selection [42], [43] to choose the next generation from the merged population (line 17). This evolutionary process continues until either the maximum number of generations or evaluations is reached. Finally, the best individual from the final population is selected to represent the MOUNTAIN's solution.

B. Solution Representation and Initialization

The solution to the MOUNTAIN problem can be represented by a tree rooted at the BS. Therefore, to solve the problem efficiently and effectively, we use node-depth encoding to represent a tree solution that has shown promising results when dealing with complex network design problems using evolutionary algorithms [40], [44], [45].

For the MOUNTAIN problem, given a connected undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{s_0, \ldots, s_K\}$, a possible solution of the problem can be represented by a spanning tree $\mathcal{T} = (\mathcal{V}, \mathcal{E}_\mathcal{T} \subset \mathcal{E})$ $(|\mathcal{E}_\mathcal{T}| = K)$. Based on the node-depth encoding, the solution is encoded by a sorted pair array with the size of $(K + 1)$, where each pair contains the node labels and its depth in the tree, i.e., $NE = [(s_0, depth_0), \ldots, (s_i, depth_i)]$, $i \in \{1, \ldots, K\}$. For instance, a complete representation of a possible solution to the MOUNTAIN problem with five physical devices in the IoT network is illustrated in Fig. 2.

To encode a routing tree, we conduct a depth-first search starting at the root node. At each node visited, we insert the pair into the data structure sequentially [46], [47]. Note that the root node s_0 has its depth $depth_0 = 0$. An example of solution encoding for the given routing tree in Fig. 2(b) is shown in Fig. 3. To decode the solution encoding to its corresponding routing tree, we traverse the array of size $K+1$ by following these steps.

1) If the current node s_i has a depth of 0, add s_i as the root node of the tree.
2) If the current node s_i has a depth greater than 0, insert s_i into the tree with the edge (s_i, s_j), where s_j is the most recently traversed node in the structure with a depth of $depth_j = depth_i - 1$.

To initialize the solution to the problem, we propose two approaches: 1) random initialization and 2) greedy initialization. For random initialization, we construct a random spanning tree with the BS as the root and the remaining nodes randomly connected to the root. For greedy initialization, the remaining nodes will be added to the tree so that the sum of the lengths of the edges of the spanning tree is the shortest, i.e., the minimum spanning tree.

Algorithm 2: Mutation Operator

Input:
- A: a parent individual submitted to mutation.
- v_k: a random vertex.
- v_p: an adjacent vertex of v_k in the input graph G.

Output: C: a child individual.

```
1 begin
2     if v_k is not ancestor of v_p in A then
3         /* Move the subtree rooted at v_k */
4         Exclude edge (v_x, v_k), where v_x is the parent of v_k in
            A ;
5         Include edge (v_p, v_k) in A.
6     end
7 end
```

C. Genetic Operators

1) Mutation: Several mutation operators have been proposed for node-depth encoding [47], [48], [49], but they require additional mechanisms to correct invalid solutions when applied to noncomplete input graphs. The recently proposed edges permutation operator (EPO) [46] addresses this issue, ensuring viable solutions without the need for correction procedures on noncomplete graphs. In this article, we adopt EPO as an efficient mutation operator for MOUNTAIN. The details of the EPO algorithm are outlined in Algorithm 2.

2) Recombination: The recombination operator creates new individuals by combining the traits of parent individuals. In this article, we employ a recombination operator based on the edges copy operator (ECO) [46], which maximizes heritability during recombination for node-depth encoding. We utilize this operator to effectively generate new solutions to the MOUNTAIN problem. Details of the ECO operator are provided in Algorithm 3. Similar to the EPO mutation, our crossover consistently produces viable individuals without requiring any additional correction mechanisms.

D. Fitness Evaluation

In our algorithms, the fitness evaluation is a process for scoring each solution based on its qualification. For the

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Algorithm 3: Recombination Operator

Input:
- A: base individual
- B: reference individual

Output: F_{AB}: generated individual by recombination of A and B.

begin
  Let s be the number of devices;
  $i \leftarrow random(\frac{s}{4}, \frac{3s}{4})$;
  Let vr be the array with size i as a random sequence of i different nodes, except the root.;
  $F_{AB} \leftarrow A$;
  foreach vr_i **do**
    if vr_i *is not the root in* A **then**
      Find a node v_p (parent of vr_i in B);
      if (vr_i *is not ancestor of* v_p *in* F_{AB}) **then**
        Perform mutation EPO on F_{AB} using the pair of nodes (vr_i, v_p);
    end
  end
end

Algorithm 4: MTEA for MOUNTAIN

Initialize population P_0 with L random individuals encoded in the USS;
t $\leftarrow$ 0;
foreach $i \in L$ **do**
  Assign a skill-factor $\tau_i = (i-1) mod L + 1$ to an individual p_i;
  Evaluate factorial cost of p_i for task τ_i only;
end
Update scalar fitness for individuals in P_0;
while *stopping criterion is not satisfied* **do**
  Offspring population $C(t) \leftarrow \emptyset$;
  while $|C(t)| < L$ **do**
    $p_a, p_b \leftarrow$ Randomize two parents from $P(t)$;
    Generate a random number *rand* between 0 and 1;
    if ($\tau_a = \tau_b$) *or* (*rand* $<$ *rmp*) **then**
      $o_a, o_b \leftarrow$ perform crossover ECO(p_a,p_b) ▷ See IV-C ;
    else
      $o_a \leftarrow$ Perform mutation EPO(p_a) ▷ See IV-C;
      $o_b \leftarrow$ Perform mutation EPO(p_b) ▷ See IV-C;
      Assign parent's skill factor for o_a, o_b;
    end
    Evaluate factorial cost of o_a, o_b;
    $C(t) \leftarrow C(t) \cup \{o_a, o_b\}$;
  end
  $P(t) \leftarrow P(t) \cup C(t)$;
  Update scalar fitness and skill factor for $P(t)$;
  $P(t+1) \leftarrow$ Select the best L individuals from $P(t)$;
  $t \leftarrow t+1$;
end

MOUNTAIN problem, the fitness value is the total energy consumption of all physical devices in the routing tree, as mentioned in (20). After obtaining the spanning tree using decoding, we calculate the objective value according to the sum of the energy consumption of all devices. Then, to check whether the solution is valid or invalid, we follow the constraints of the MOUNTAIN, i.e., (13)–(17). If one of the constraints is not satisfied, the fitness value of positive infinity will be assigned as a penalty.

E. MTEA: Proposed Multitask Evolutionary Algorithm

In recent years, evolutionary multitasking has been successfully applied to solve various optimization problems simultaneously [39], [40], [50]. In this section, we introduce a novel MTEA, which builds upon the evolutionary multitasking optimization. MTEA is designed to address different instances of the MOUNTAIN problem concurrently, allowing for knowledge transfer among these instances. This approach is particularly effective when optimizing one problem can facilitate the optimization of another.

In our algorithm, we treat each MOUNTAIN instance as an optimization task and solve these instances simultaneously. For instance, we can use MTEA to address two MOUNTAIN instances with 10 and 15 devices at the same time. To achieve this, we encode the search spaces for each task into a unified space. The genetic operators are then applied within this space, aiming to reduce computational costs while taking advantage of knowledge transfer between different tasks. The framework of our MTEA is presented in Algorithm 4.

1) Preliminaries of Multitask Evolutionary Algorithm: The objective of multitask evolutionary algorithms, e.g., MFEA [32], is to tackle multiple optimization problems (or tasks) at the same time. For simplification, we can treat each task as a minimization optimization problem. Thus, the main goal is to identify a set of optimal solutions that meet the requirements of all tasks. This can be formulated as follows:

$$(X_1^*, X_2^*, \ldots, X_\Gamma^*) = \underset{s.t. \forall X_\gamma \in \Omega_\gamma}{\arg\min} \ (f_1(X), f_2(X), \ldots, f_\Gamma(X))$$

where Ω_γ and f_γ represent the decision search space and objective function of the γth task, denoted as T_γ, respectively.

MFEA addresses multiple optimization tasks simultaneously using a unified search space (USS). Each individual solution X^i in the population is characterized by four key metrics.

1) *Factorial Cost:* $\Psi_\gamma^i \in \mathbb{R}$, representing the fitness value of X^i for task T_γ, calculated as $f_\gamma(X_\gamma^i)$ after decoding X^i into its task-specific representation X_γ^i.
2) *Factorial Rank:* $r_\gamma^i \in \mathbb{N}$, indicating the rank of X^i for task T_γ within the population when sorted by its factorial cost Ψ_γ^i in ascending order.
3) *Scalar Fitness:* φ^i, defined as the reciprocal of the best factorial rank across all tasks: $\varphi^i = 1/\min_k r_\gamma^i$. A larger scalar fitness signifies better overall performance.
4) *Skill Factor:* τ^i, identifying the task where X^i performs best, determined by $\tau^i = \arg\min_k r_\gamma^i$.

These metrics enable MFEA to evaluate and select individuals effectively within a multitasking optimization framework.

2) Unified Solution Representation: We consider each MOUNTAIN instance as a task in the multitask optimization setting, denoted T_γ ($\gamma = (1, \ldots, \Gamma)$). For each task, the corresponding input graph is represented by $\mathcal{G}_\gamma = (\mathcal{V}_\gamma, \mathcal{E}_\gamma)$ with vertex set $\mathcal{V}_\gamma$ and edge set $\mathcal{E}_\gamma$. We denote the unified

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graph of all tasks as $\mathcal{G}^u = (\mathcal{E}^u, \mathcal{V}^u)$, where $\mathcal{V}^u = \mathcal{V}_1 \cup \mathcal{V}_2 \cup \cdots \cup \mathcal{V}_\Gamma$ and $\mathcal{E}^u = \mathcal{E}_1 \cup \mathcal{E}_2 \cup \cdots \cup \mathcal{E}_\Gamma$.

To encode the solutions for all tasks at once, we use $NE = [(s_0, depth_0), \ldots, (s_i, depth_i)], i \in \{1, \ldots, |\mathcal{V}^u| - 1\}$ as the unified solution encoding. This encoding represents a random routing tree in the unified graph $\mathcal{G}^u$. For example, when we use MTEA to solve two MOUNTAIN instances with 5 and 10 devices (including the BS), the unified encoding NE has a length of 10.

To decode the solution for each task from the unified encoding, we propose a new decoding method that always constructs a valid routing tree for each MOUNTAIN instance from the unified search space. We denote the random routing tree of the graph $\mathcal{G}_\gamma = (\mathcal{V}_\gamma, \mathcal{E}_\gamma)$ for the task T_γ as $\mathcal{T} = (\mathcal{V}_\mathcal{T}, \mathcal{E}_\mathcal{T})$. At initialization, we set $\mathcal{V}_\mathcal{T} = \emptyset$ and $\mathcal{E}_\mathcal{T} = \emptyset$. The decoding process to construct $\mathcal{T}$ is performed by traverse NE with the following steps.

1) If the $depth_i$ of the current node s_i is 0, insert the node s_i as the root to the tree $\mathcal{T}$ without any edges, i.e., $\mathcal{V}_\mathcal{T} = \{s_i\}$ and $\mathcal{E}_\mathcal{T} = \emptyset$, and remove $(s_i, depth_i)$ from NE.
2) If the $depth_i$ of the current node s_i is larger than 0 and $s_i \in \mathcal{V}_\gamma$, we find the node $s_j \in \mathcal{V}_\mathcal{T}$ that satisfies $depth_j < depth_i$ and $|i - j|$ being minimal. If edge $(s_i, s_j) \in \mathcal{E}_\gamma$, then s_i and (s_i, s_j) are added to $\mathcal{T}$ and $(s_i, depth_i)$ is removed from NE. In contrast, the depth of node s_i is updated to $depth_i + 1$.
3) If $|\mathcal{V}_\mathcal{T}| < |\mathcal{V}_\gamma|$ then repeat the Step 2.

We prove that the decoding process consistently creates a random routing tree for each MOUNTAIN instance from the unified search space of MTEA, as shown by Propositions 1–3.

Proposition 1: Let $\mathcal{T}$ be a tree represented by a node-depth encoding, rooted at a node s_0 with $depth_0 = 0$. Denote the set of nodes and edges of $\mathcal{T}$ as $\mathcal{V}_\mathcal{T}$ and $\mathcal{E}_\mathcal{T}$, respectively. Then, for any node $s_i \in \mathcal{V}_\mathcal{T}$, the depth of s_i, denoted as $depth_i$, satisfies

$$\text{depth}_i \leq |\mathcal{V}_\mathcal{T}| - 1.$$

Proof: The depth $depth_i$ of a node $s_i \in \mathcal{T}$ is defined as the number of edges in the unique path from the root node s_0 to s_i. Since $\mathcal{T}$ is a tree, it is connected and acyclic, and every pair of nodes in $\mathcal{T}$ is connected by exactly one unique path. The maximum depth in $\mathcal{T}$ is attained in the longest path from the root s_0 to a leaf node. In the extreme case where $\mathcal{T}$ forms a path graph (a single line of nodes), the depth of the farthest node from the root is $|\mathcal{V}_\mathcal{T}| - 1$, which is the maximum number of edges in $\mathcal{T}$. For all other cases, the depth of any node s_i is strictly less than $|\mathcal{V}_\mathcal{T}| - 1$. Hence, for all $v_i \in \mathcal{V}_\mathcal{T}$

$$depth_i \leq |\mathcal{V}_\mathcal{T}| - 1.$$

■

Proposition 2: Let NE^m represent the unified encoding, and $\mathcal{T}^m$ the tree resulting from performing step 2 in the decoding process m times. At every iteration m, there is at least one node $s_i \in NE^m$ such that s_i can be added to the current tree $\mathcal{T}^m$.

Proof: We aim to demonstrate that there exists at least one node s_i in NE^m that can be added to the tree $\mathcal{T}^m$. Assume, for the sake of contradiction, that no node s_i can be added to $\mathcal{T}^m$. This would imply that $depth_i^m \leq depth_j^m$ for all nodes s_j. If s_i is not added to the tree at the mth iteration, $depth_i^m$ increases by one. Consequently, $|\mathcal{V}_\mathcal{T}^m| \leq m$ and $m \leq depth_i^m$, leading to $|\mathcal{V}_\mathcal{T}^m| \leq depth_i^m \leq depth_j^m$ (*). According to Proposition 1, we have $depth_j^m \leq |\mathcal{V}_\mathcal{T}^m| - 1$ for all $s_j \in \mathcal{V}_\mathcal{T}^m$, thus $depth_j^m < |\mathcal{V}_\mathcal{T}^m|$ (**). Combining (*) and (**), we arrive at a contradiction. Therefore, our assumption is false, and there must exist at least one node s_i in NE^m that can be added to $\mathcal{T}^m$, completing the proof. ■

Proposition 3: To construct a valid routing tree $\mathcal{T}$ for instance T_γ from NE, step 2 is performed at most $|\mathcal{V}_\mathcal{T}| - 1 = |\mathcal{V}^u| - 1$ times where $|\mathcal{V}^u|$ is the maximum number of devices among MOUNTAIN instances.

Proof: According to Proposition 2, at each iteration of step 2, at least one node s_i is added to the tree $\mathcal{T}^m$. Since a spanning tree $\mathcal{T}$ with $|\mathcal{V}^u|$ nodes already has a root node s_0, step 2 terminates after at most $|\mathcal{V}^u| - 1$ iterations. This guarantees that the decoding process completes with a valid routing tree $\mathcal{T}$ for the task T_γ. ■

F. Complexity Analysis

In this section, we analyze the time complexity of our proposed algorithms, STEA and MTEA, designed for MOUNTAIN. Let L denote the number of individuals in the population, and G represent the number of generations in our algorithms.

In STEA, the algorithm's time complexity is determined by two primary stages, including population initialization in the first generation and the reproduction process over the subsequent G generations. During population initialization, we generate L random spanning trees by the PrimRST algorithm [51] and encode them by DFS with the time complexity of $\mathcal{O}(L \times |\mathcal{E}|)$ and $\mathcal{O}(L \times (|\mathcal{E}| + |\mathcal{V}|))$, respectively. Next, we evaluate the fitness for individuals by decoding and calculating the objective value with time complexity of $\mathcal{O}(L \times |\mathcal{V}|)$. Therefore, the overall time complexity of initialization is $\mathcal{O}(L \times (|\mathcal{E}| + |\mathcal{V}|)$. During offspring reproduction, we generate L new individuals by recombination and mutation with the time complexity of $\max(\mathcal{O}(L \times |\mathcal{V}|^2), \mathcal{O}(L \times |\mathcal{V}|)) = \mathcal{O}(L \times |\mathcal{V}|^2)$. We then evaluate fitness for offspring individuals with the time complexity of $\mathcal{O}(L \times |\mathcal{V}|)$. For selection to the next generation, its time complexity is $\mathcal{O}(L \times \log L)$. Therefore, the overall time complexity of reproduction in G generations is $\mathcal{O}(G \times L \times (|\mathcal{V}|^2 + \log L))$ as $|\mathcal{V}|^2 >> |\mathcal{V}|$. To sum up, the time complexity of our STEA is: $\mathcal{O}(\text{STEA}) = \mathcal{O}(L \times (|\mathcal{E}| + |\mathcal{V}|)) + \mathcal{O}(G \times L \times (\mathcal{V}|^2 + \log L)) = \mathcal{O}(L \times (|\mathcal{E}| + G \times (|\mathcal{V}|^2 + \log L)))$.

In MTEA, the time complexity of this algorithm also comes from two main stages that are similar to STEA. However, the difference to STEA lies in the solution decoding method in a multitask setting. According to Proposition 3, step 2 in the decoding needs to be repeated at most $|\mathcal{V}^u| - 1$ times to construct the valid routing tree $\mathcal{T}$ for a task. At each iteration m, it has time complexity of $\mathcal{O}(m \times (|\mathcal{V}^u| - m))$ when traverse $\mathcal{T}^m$ and NE^m. Therefore, the time complexity of the decoding process is $\sum_{m=1}^{|\mathcal{V}^u|} m \times (|\mathcal{V}^u| - m) = (1/6) \times (|\mathcal{V}^u| - 1) \times |\mathcal{V}^u| \times (|\mathcal{V}^u| + 1) = \mathcal{O}(|\mathcal{V}^u|^3)$. Following the steps similar to STEA, we obtain the time complexity of MTEA as follows:

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$\mathcal{O}(\text{MTEA}) = \mathcal{O}(L\times(|\mathcal{E}|+|\mathcal{V}^3|)) + \mathcal{O}(G\times L\times(\mathcal{V}|^3+\log L)) = \mathcal{O}(L \times (|\mathcal{E}| + G \times (|\mathcal{V}|^3 + \log L))).$

V. Performance Evaluation

In this section, we present the simulation results of the proposed algorithms and other baselines. We generate synthetic datasets by uniformly distributing K IoT devices within a 200 m × 200 m area. The BS is fixed at the center of the area [coordinates (100, 100)], and a subset of M devices with relay capabilities is strategically positioned to maintain connectivity to the BS. Device locations are sampled using a uniform random distribution, ensuring that our datasets reflect varied network densities. Varying the number of devices K in the fixed area allows us to simulate different network densities. Higher densities provide more routing options, potentially reducing energy consumption as devices can relay data more efficiently. However, increased density may also lead to higher interference and congestion, which can impact transmission delays. All other constraint parameters in (15)–(19) are also consistent with those in [7], ensuring fairness in comparison and reproducibility. For example, the accuracy threshold $A_n = 0.6$ and latency limit $L_n = 0.1$ s were shown to effectively balance data fidelity and communication cost in digital twin-enabled IoT systems. These values reflect realistic expectations for digital twin accuracy and responsiveness in smart environments. Additionally, the sensitivity parameter α used in the accuracy function is likewise inherited from [7] and [52], where it was empirically validated to capture the tradeoff between data completeness and energy efficiency.

We compare the performance of our proposed evolutionary algorithms, namely, STEA and MTEA, with four baselines, including a random algorithm (labeled as "Random"), a greedy algorithm (labeled as "Greedy"), a strong nature-inspired metaheuristic for global optimization, namely, AVOA [29], and an efficient routing protocol based on Genetic Algorithm (labeled as "GA") [23], [53], [54]. The Random algorithm constructs a random tree, and the Greedy algorithm constructs a minimum-spanning tree, i.e., Prim's and Kruskal's, that satisfies the constraints of the problem. The AVOA and GA search the tree solutions based on the network random-key encoding [55], and GA utilizes appropriate genetic operators, including Simulated Binary Crossover and Polynomial Mutation [39]. The details of simulation parameters of the system [7] and proposed algorithms are given in Table II.

First, to evaluate the convergence of the proposed algorithms, we compare the convergence trends of STEA and MTEA with those of the Random, Greedy, GA, and AVOA algorithms across multiple iterations. Fig. 4 illustrates the obtained objective value (total energy consumption) over generations for each algorithm. The results demonstrate that MTEA converges within approximately 30 iterations, achieving the best solution quality among all algorithms. STEA, while requiring around 40 iterations to converge, still outperforms the baseline algorithms, including Random, Greedy, GA, and AVOA. These findings indicate that the proposed algorithms not only converge within a reasonable number of iterations but also consistently yield superior solutions compared to the baselines. This results from the algorithms' ability to use effective solution representation and genetic operators, which allows for efficient exploration and exploitation of the solution space, ultimately producing high-quality solutions for the MOUNTAIN problem. Compared to our single-task optimization approach, MTEA converges both faster and more effectively than STEA. This improvement is attributed to our multitask optimization strategy with the new unified solution representation and effective solution decoding method, which

TABLE II
Simulation Parameters

Description	Parameters	Value
Time frame	T	1 second
Number of devices	K	5, 10, 15, 20
Number of relays	M	2, 3, 4, 5
Maximum number of resource blocks	K_0	2, 3, 4, 5
Number of time slots that device and server must have at least one communication	τ	3
Constant	β_k	1/3
Area size	H	200 m × 200 m
Data of each device	D_{nk}	300 kbits
Time slot	N	10
Bandwidth	B	1 MHz
Maximum transmit power	P_k	1 dBm
Accuracy requirement	A_n	0.6
Maximum latency permission	L_n	0.1 second
Noise power spectral density	N_0	-174 dBm/Hz
Chanel gain	h_{nk}	pathloss (code)
Waterfall threshold	m	0.023 dB
Communication range	r_c	100
Population size	L	$K \times 5$
Maximum number of generations	G	$K \times 6$
Crossover rate	p_c	0.95
Mutation rate	p_m	0.6
Probability of random mating	rmp	0.7

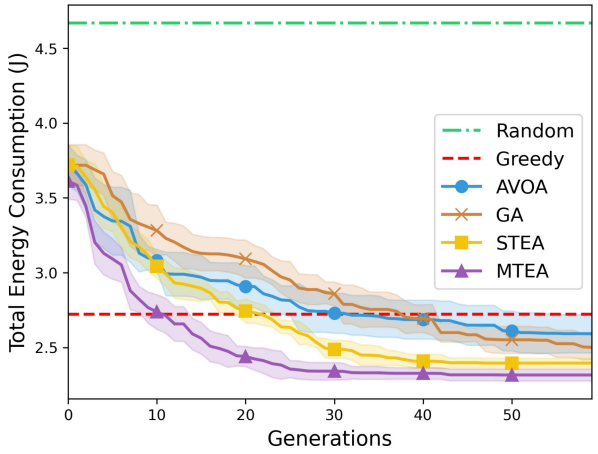


Fig. 4. Convergence trend of all algorithms in the network with ten physical devices.

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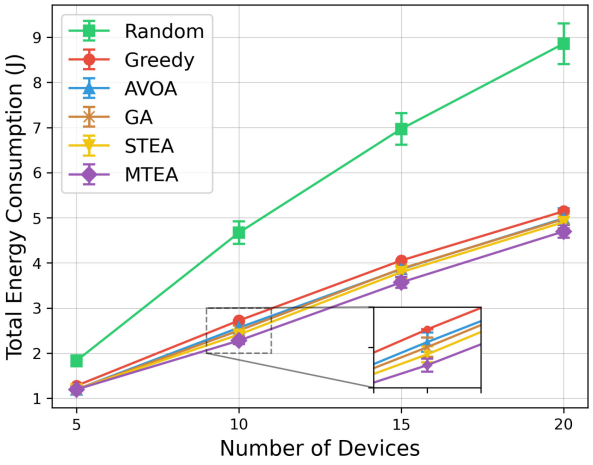


Fig. 5. Total energy consumption versus the number of physical devices.

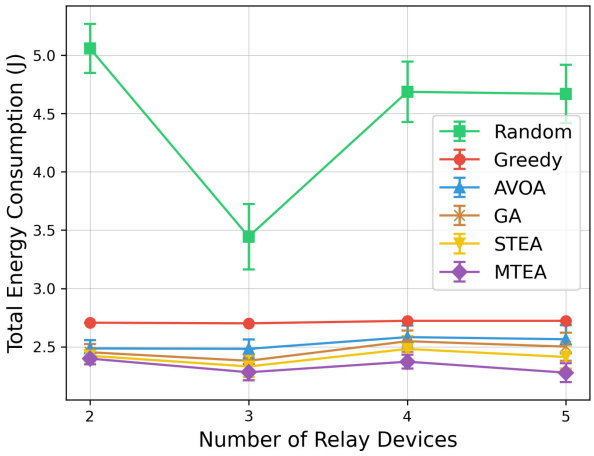


Fig. 6. Total energy consumption versus the number of relay devices in the network with ten physical devices.

enhances MTEA's performance throughout the parallel evolutionary process by facilitating implicit knowledge exchange among MOUNTAIN instances when solving at the same time. Furthermore, MTEA leverages the robustness of the evolutionary multitask optimization paradigm, which previous studies have demonstrated to be more effective than single-task approaches [32], [33], [37], [38], [39], [40].

Second, we evaluate the performance of the proposed algorithms in comparison to the baseline methods in terms of the total energy consumption of the network (objective value). Fig. 5 illustrates the energy consumption of the network corresponding to different numbers of physical devices, specifically 5, 10, 15, and 20 devices. It can be observed that the network's energy consumption increases as the number of devices increases. In particular, as the number of devices grows, the gaps in energy consumption between the proposed algorithm MTEA and the remaining algorithms (Greedy, AOVA, GA, and STEA) gradually widen, and the gap with the Random algorithm increases even more rapidly. Although the number of devices increases and the network's energy consumption also rises, our MTEA can find the best routing solutions with smaller total energy consumption among those obtained by all algorithms. Overall, our MTEA consistently finds the best objective value compared to the other methods across different numbers of devices.

Third, we evaluate the impact of factors, such as the number of relay-capable devices in the network and the maximum number of resource blocks on total energy consumption. Fig. 6 illustrates the effect of increasing the number of relay devices on energy consumption. The proposed algorithms, STEA and MTEA, consistently achieve the lowest energy consumption across all scenarios, demonstrating their superior performance compared to the baseline algorithms (Greedy, GA, and AVOA) and the Random algorithm. As the number of relay devices increases, the energy consumption of STEA and MTEA exhibits a slight yet consistent decrease, reflecting their ability to effectively leverage improved connectivity. In contrast, the energy consumption of the Random algorithm fluctuates significantly, while that of the Greedy algorithm remains unchanged. The results obtained by AVOA do not

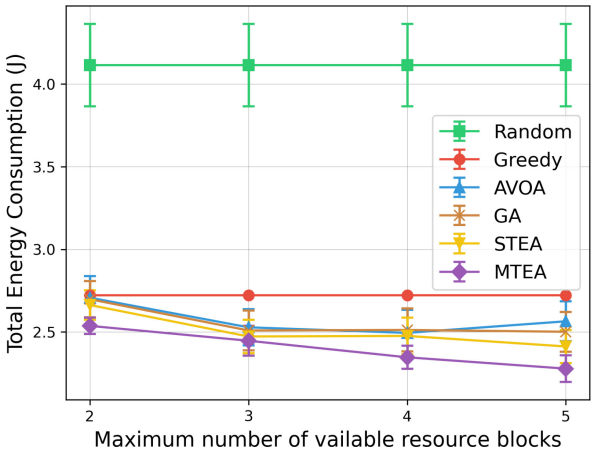


Fig. 7. Total energy consumption obtained by our method versus the maximum number of available resource blocks in the network with ten physical devices.

decrease consistently, as AVOA fails to explore diverse and near-optimal solutions due to redundant solution encoding and ineffective search operators. These results highlight that the proposed algorithms efficiently utilize the additional relay devices to enhance connectivity, reduce energy consumption, and outperform other approaches.

Fig. 7 illustrates the relationship between total energy consumption and the maximum number of available resource blocks for different algorithms. The Random and Greedy algorithms exhibit the highest energy consumption, with no improvement across different resource block configurations, reflecting their lack of effective optimization strategies. AVOA and GA perform moderately better but are still outperformed by the proposed algorithms. In contrast, STEA and MTEA demonstrate significant reductions in energy consumption, leveraging additional resource blocks more effectively. Notably, MTEA achieves the lowest energy consumption across all configurations, demonstrating its superior efficiency in utilizing available resources. These results highlight the effectiveness of the proposed algorithms in reducing energy consumption while leveraging additional resource blocks.

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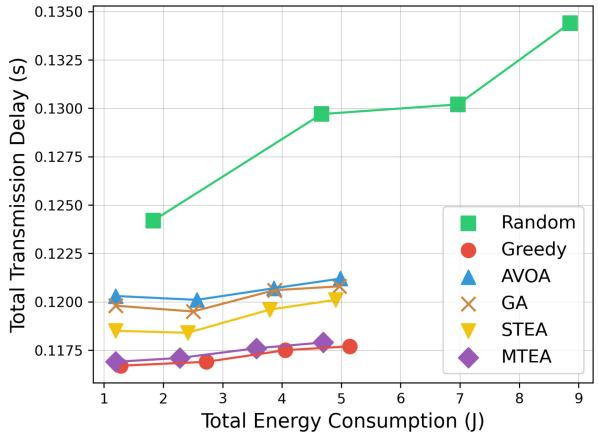


Fig. 8. Correlation between the total energy consumption and transmission delay. Each curve has four points with the number of devices $K = [5, 10, 15, 20]$ (from left to right).

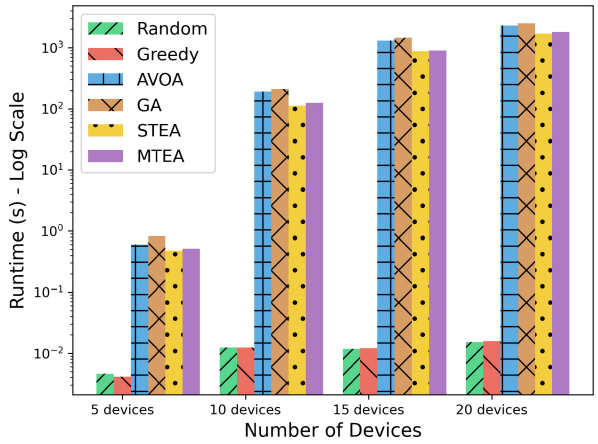


Fig. 9. Running time of algorithms.

Fig. 8 illustrates the correlation between total energy consumption and transmission delay, with results plotted for $K = \{5, 10, 15, 20\}$. The proposed MTEA algorithm consistently achieves the lowest energy consumption and transmission delay across all scenarios, demonstrating its efficiency and scalability. Compared to STEA, MTEA exhibits superior performance, particularly under higher device counts, where it maintains minimal increases in delay and stable energy consumption. In contrast, the Random algorithm shows erratic behavior, with significantly higher energy consumption and sharply increasing delays as the number of devices grows, highlighting its inefficiency. While the Greedy algorithm achieves slightly lower delays at specific points, it sacrifices energy efficiency, resulting in higher overall energy consumption compared to MTEA and STEA. AVOA and GA perform moderately but are outperformed by MTEA in both metrics. These insights confirm MTEA's robustness and effectiveness in balancing energy consumption and delay, making it the most reliable method for resource-efficient network optimization.

Finally, we assess the running time of the algorithms, as shown in Fig. 9. The Random and Greedy algorithms exhibit similar running times and produce solutions relatively quickly compared to our proposed algorithms. However, when considering the tradeoff between execution time and solution quality, our proposed algorithms, STEA and MTEA, consistently deliver significantly better energy consumption solutions than the Random and Greedy approaches, all within an acceptable time frame. It can be explained that the reason is that the iterative refinement procedure in STEA and MTEA requires additional computation time, but remains within reasonable and acceptable limits compared to Random and Greedy algorithms. Significantly, our algorithms require less time and offer better solutions than AVOA and GA, which use netkeys encoding and suffer from the issue of redundant solution representation in the search space.

VI. Conclusion

In this article, we addressed the MOUNTAIN problem by introducing two novel evolutionary algorithms—STEA and MTEA—that optimize multihop routing in IoT-based digital twin systems. Our approaches minimize energy consumption while balancing digital twin accuracy and communication costs through specialized encoding and genetic operators. Extensive experiments demonstrate that both algorithms significantly outperform AVOA, Greedy, and Random baselines in energy efficiency and network stability. The MTEA algorithm achieves particularly robust performance through its knowledge-sharing mechanism, contributing to sustainable IoT infrastructures that support high-fidelity digital twins. Our study has limitations, including the use of synthetic datasets with uniform device distributions and static network conditions. Future work will extend these algorithms to heterogeneous, dynamic environments and explore cross-layer optimization approaches combining our routing strategies with physical layer techniques [56], [57] to enhance the efficiency and reliability of IoT-based digital twin systems.

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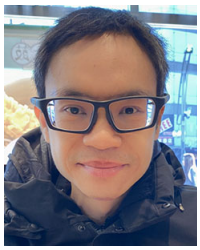
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